Composite Wing Structural Design
Preliminary Design Review (PDR++)

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Chapter 1

Problem Definition and Scope

1.1 Background

The structural design of aircraft wings requires a careful balance between aerodynamic efficiency, structural integrity, and weight minimization. In the preliminary design phase, simplified analytical models are typically employed to establish first-order estimates of loads, stiffness requirements, and geometric parameters.

This report focuses on the conceptual structural design of a composite wing for a light aircraft configuration. The intent is not to produce a finalized design, but rather to develop a consistent framework that links aerodynamic loading with structural response and material behavior.

1.2 Objective of the Study

The primary objective of this work is to determine the structural characteristics of a composite wing-box configuration under representative loading conditions. This involves identifying the governing loads, estimating the required stiffness, and sizing the primary structural elements accordingly.

The study aims to answer the following key questions:

- What are the spanwise distributions of shear force and bending moment acting on the wing?
- What level of bending stiffness is required to satisfy deflection constraints?
- How can composite laminate theory be used to determine equivalent material properties?
- What should be the dimensions of the spar caps and wing-box structure?
- What spacing of ribs is required to ensure structural stability?

1.3 Scope of Work

The analysis carried out in this report is limited to a first-order structural design. The following aspects are included:

- Aerodynamic load modeling using an idealized spanwise lift distribution
- Determination of shear force, bending moment, and torsional moment
- Evaluation of bending stiffness requirements based on tip deflection constraints
- Application of Classical Laminate Theory (CLT) to obtain equivalent material properties
- Preliminary sizing of spar caps and wing-box geometry
- Estimation of torsional rigidity of the wing box
- Determination of rib spacing based on shear-driven stability considerations

1.4 Assumptions and Simplifications

In order to keep the analysis tractable at the preliminary design stage, several simplifying assumptions are adopted:

- The wing is modeled as a straight, untwisted beam with no taper or dihedral
- The lift distribution is approximated as elliptical
- The wing structure is represented using classical beam theory
- Material behavior is represented using Classical Laminate Theory
- Load transfer between components is assumed to be ideal

These assumptions allow for analytical expressions to be used, enabling rapid estimation of structural parameters while maintaining reasonable physical accuracy.

Chapter 2

Aircraft and Wing Configuration

2.1 Overview

The structural design of a wing is strongly influenced by the overall aircraft configuration and its geometric characteristics. At the preliminary design stage, simplified representations of the aircraft are used to establish consistent loading conditions and structural requirements.

This chapter defines the fundamental parameters of the aircraft and the geometric properties of the wing that are used throughout the analysis.

2.2 Aircraft-Level Parameters

The aircraft considered in this study is representative of a light, low-speed platform. For such aircraft, structural design is typically governed by maneuver loads rather than compressibility or high-speed aerodynamic effects.

The key parameters used in the present analysis are listed in Table 2.1.

Table 2.1: Primary Aircraft Parameters

Maximum Take-Off Mass (MTOW)	580 kg
Weight W	5680 N
Cruise Speed V_c	50 m/s
Stall Speed V_s	23 m/s

The weight of the aircraft is obtained using the relation:

$$W = mg \tag{2.1}$$

where m is the maximum take-off mass and g is the gravitational acceleration.

2.3 Wing Planform Geometry

The wing is modeled using a simplified geometric representation in order to enable analytical treatment of loads and structural response. The planform is assumed to be rectangular, with

no taper or sweep.

The geometric parameters of the wing are summarized in Table 2.2.

Table 2.2: Wing Geometric Parameters

Wingspan b	11 m
Semi-span $s = b/2$	5.5 m
Wing Area S	13.5 m^2
Aspect Ratio AR	9
Mean Aerodynamic Chord c	1.23 m

2.4 Design Interpretation of Geometry

The chosen geometric parameters reflect a balance between aerodynamic efficiency and structural simplicity.

- A moderate aspect ratio ($AR = 9$) provides improved aerodynamic efficiency while keeping bending loads within manageable limits.
- The absence of sweep eliminates aeroelastic coupling effects such as bend-twist interaction, simplifying structural analysis.
- The rectangular planform allows for analytical expressions of load distributions, avoiding the need for numerical approximations at this stage.

2.5 Semi-Span Representation

All structural calculations in this study are performed on a single wing (semi-span). This is justified by the symmetry of the aircraft configuration.

The semi-span is defined as:

$$s = \frac{b}{2} = 5.5\,\text{m} \tag{2.2}$$

The use of semi-span representation simplifies the formulation of shear force and bending moment distributions, while still capturing the complete structural behavior of the wing.

2.6 Wing Box Geometric Assumptions

For structural analysis, the wing is idealized as a wing-box structure embedded within the airfoil. The wing box is the primary load-carrying component and consists of:

- Front and rear spars
- Upper and lower skins

- Internal ribs for load redistribution

The following simplifying assumptions are made:

- The wing box occupies a fixed fraction of the chord
- The structure is symmetric about the mid-plane
- The spar caps carry bending loads as axial forces
- The spar webs carry shear loads
- The skins contribute to torsional rigidity

2.7 Preliminary Geometric Estimates

Based on typical proportions for light aircraft wings, the following preliminary estimates are adopted:

- Wing-box width $\approx 40\%$ to 50% of chord
- Spar height determined from airfoil thickness
- Uniform structural layout along the span

These estimates will be refined in later chapters once stiffness and load requirements are established.

2.8 Wing Planform Characteristics

The wing is modeled as a straight, untwisted configuration to simplify structural analysis.

Wingspan b	11 m
Semi-span s	5.5 m
Wing Area S	13.5 m^2
Aspect Ratio	9
Mean Chord c	1.23 m

This simplified geometry allows direct application of classical beam theory for load estimation.

Chapter 3

Airfoil Characteristics and Structural Idealization

3.1 Introduction

The aerodynamic shape of the wing plays a critical role in determining both its aerodynamic performance and structural feasibility. In particular, the airfoil geometry directly influences the available structural depth, which governs bending stiffness and load-carrying capability.

This chapter presents the selected airfoil and develops the structural idealization of the wing in terms of a simplified wing-box model.

3.2 Airfoil Selection

The airfoil selected for this study is the NACA 2412 airfoil. This airfoil is widely used in low-speed aircraft due to its favorable lift characteristics and predictable aerodynamic behavior.

The key properties of the selected airfoil are summarized in Table 3.1.

Table 3.1: Airfoil Properties

Airfoil Type	NACA 2412
Thickness Ratio (t/c)	0.12
Maximum Lift Coefficient ($C_{L,\max}$)	≈ 1.5
Typical Reynolds Number Range	10^6 – 10^7

3.3 Structural Significance of Airfoil Thickness

The thickness-to-chord ratio ($t/c = 0.12$) plays a crucial role in structural design. The available structural depth determines the separation between the upper and lower load-carrying members (spar caps), which directly affects bending stiffness.

The maximum airfoil thickness at the root is estimated as:

$$t_{\max} = 0.12 \times c \tag{3.1}$$

Substituting the chord value:

$$t_{\max} = 0.12 \times 1.23 \approx 0.15\,\mathrm{m} \tag{3.2}$$

This thickness provides sufficient depth for accommodating a structurally efficient wing-box.

3.4 Airfoil Thickness Distribution

The variation of airfoil thickness along the chord is described using the standard NACA 4-digit thickness distribution:

$$y_t(x) = 5t\left(0.2969\sqrt{x} - 0.126x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4\right) \tag{3.3}$$

where:

- x is the normalized chordwise coordinate,
- t is the maximum thickness ratio.

This expression is used to determine the local airfoil thickness at any chordwise position.

3.5 Wing-Box Structural Idealization

For structural analysis, the wing is modeled as a closed wing-box structure embedded within the airfoil. This is a standard representation used in aircraft structures due to its efficiency in carrying combined bending, shear, and torsional loads.

The wing-box consists of the following primary components:

- **Spar Caps:** Carry axial forces generated by bending moments
- **Spar Webs:** Transfer shear forces along the span
- **Skins:** Provide torsional rigidity and contribute to load distribution
- **Ribs:** Maintain cross-sectional shape and redistribute localized loads

3.6 Load Path in the Wing Structure

The load transfer mechanism within the wing box can be summarized as follows:

- The aerodynamic lift generates a distributed load along the span
- This load produces a bending moment that is resisted by axial forces in the spar caps

- Shear forces are carried primarily by the spar webs
- Torsional moments are resisted by shear flow in the closed wing-box section

This separation of load paths allows different structural components to be designed based on their dominant loading modes.

3.7 Spar Placement

The location of spars within the airfoil is determined based on both structural and aerodynamic considerations.

Typical placement is assumed as:

- Front spar: located at approximately 25% of chord
- Rear spar: located at approximately 65% of chord

These locations ensure that:

- The spars lie within the region of maximum airfoil thickness
- A closed section is formed for torsional resistance
- Sufficient spacing exists between spars to maximize bending efficiency

3.8 Determination of Spar Heights

The heights of the front and rear spars are obtained from the airfoil thickness distribution:

$$h_f = 2y_t\left(\frac{x_f}{c}\right), \quad h_r = 2y_t\left(\frac{x_r}{c}\right) \tag{3.4}$$

where:

- x_f and x_r are the front and rear spar locations,
- c is the chord length.

These heights define the effective distance between the upper and lower spar caps and are critical in determining bending stiffness.

3.9 Wing Box Dimensions

The effective width of the wing box is defined as:

$$b_{\text{box}} = x_r - x_f \tag{3.5}$$

This width, along with the spar heights, defines the cross-sectional geometry of the wing box.

3.10 Structural Modeling Assumptions

The following assumptions are made for the structural idealization:

- The wing box behaves as a thin-walled closed section
- The material is treated as orthotropic due to composite construction
- The structure is symmetric about the mid-plane
- Local stress concentrations are neglected in preliminary sizing

These assumptions enable analytical expressions to be used for stiffness and load calculations.

3.11 Summary

This chapter establishes the geometric and structural representation of the wing. The airfoil provides the physical constraints for structural placement, while the wing-box idealization defines how loads are carried.

The resulting geometry forms the foundation for aerodynamic load calculations and structural sizing in subsequent chapters.

Chapter 4

Loading Conditions

4.1 Design Load Factors

The structure is sized based on maneuver loading conditions:

Limit Load Factor	+3.8
Negative Load Factor	-1.52
Ultimate Load Factor	1.5 × limit

4.2 Total Lift Estimation

$$L = nW \tag{4.1}$$

$$L_{\text{limit}} = 3.8 \times 5680 = 21584\,\text{N} \tag{4.2}$$

$$L_{\text{ultimate}} = 1.5 \times L_{\text{limit}} = 32376\,\text{N} \tag{4.3}$$

Chapter 5

Loading Conditions and Design Envelope

5.1 Introduction

The structural design of an aircraft wing is governed primarily by the loads it experiences during operation. These loads arise due to aerodynamic forces, maneuvering conditions, and safety requirements imposed by design standards.

This chapter defines the loading conditions used for the structural sizing of the wing. The focus is on identifying the governing load cases and translating aircraft-level parameters into wing-level forces.

5.2 Design Load Factors

Aircraft structures are designed to withstand loads corresponding to specified maneuver conditions. These conditions are expressed in terms of load factors, defined as:

$$n = \frac{L}{W} \tag{5.1}$$

where L is the lift force and W is the weight of the aircraft.

For the present analysis, the following load factors are considered:

- Positive limit load factor: $n_{\text{lim}} = +3.8$
- Negative limit load factor: $n_{\text{neg}} = -1.52$
- Ultimate load factor: $n_{\text{ult}} = 1.5 \times n_{\text{lim}} = 5.7$

The ultimate load factor incorporates a factor of safety to ensure structural reliability under extreme conditions.

5.3 Lift Estimation Under Maneuver Conditions

The total lift generated by the wing under a given load factor is:

$$L = nW \tag{5.2}$$

Substituting the aircraft weight:

$$L_{\text{limit}} = 3.8 \times 5680 = 21584\,\text{N} \tag{5.3}$$

$$L_{\text{ultimate}} = 1.5 \times 21584 = 32376\,\text{N} \tag{5.4}$$

These values represent the primary design loads for the wing structure.

5.4 Load Distribution Between Wings

For a symmetric aircraft configuration, the total lift is shared equally between the left and right wings. Therefore, the load acting on a single wing (semi-span) is:

$$L_{\text{semi}} = \frac{L}{2} \tag{5.5}$$

Thus,

$$L_{\text{semi, limit}} = 10792\,\text{N} \tag{5.6}$$

$$L_{\text{semi, ultimate}} = 16188\,\text{N} \tag{5.7}$$

All subsequent structural calculations are performed using the semi-span loads.

5.5 Assumed Load Distribution

Although the wing planform is rectangular, the lift distribution is approximated as elliptical. This assumption is commonly used because:

- It represents an aerodynamically efficient loading condition
- It simplifies analytical integration
- It provides conservative estimates for structural design

The elliptical distribution ensures smooth variation of loads along the span and avoids unrealistic discontinuities.

5.6 Nature of Structural Loading

The distributed aerodynamic load acting on the wing results in three primary structural effects:

- **Shear Force:** Arises due to the cumulative effect of lift along the span
- **Bending Moment:** Generated due to the moment arm of the lift force about the root
- **Torsional Moment:** Caused by the offset between aerodynamic center and shear center

These load components form the basis for structural sizing.

5.7 Critical Load Case Selection

Among the various loading scenarios, the positive maneuver limit load case is typically the governing condition for preliminary structural sizing. This is because:

- It produces the highest upward bending moment
- It results in maximum tensile and compressive stresses in the spar caps
- It defines the primary stiffness requirement for the wing

The ultimate load case is used to verify structural safety margins, while the limit load case is used for stiffness-driven design.

5.8 Design Philosophy for Loading

The approach adopted in this study is based on the following principles:

- Use limit loads to determine structural stiffness requirements
- Use ultimate loads to evaluate strength and safety margins
- Assume smooth load distributions to enable analytical solutions
- Focus on global structural response rather than local effects

This ensures a consistent and physically meaningful basis for preliminary design.

5.9 Summary

This chapter establishes the loading framework used for the structural design of the wing. The aircraft-level parameters have been translated into wing-level forces, and the governing load cases have been identified.

These loads are used in the next chapter to derive spanwise distributions of shear force, bending moment, and torsional moment.

Chapter 6

Aerodynamic Load Modeling

6.1 Introduction

Having established the governing loading conditions, the next step is to determine how these loads are distributed along the span of the wing. The spanwise variation of lift directly influences the shear force, bending moment, and torsional moment distributions, which in turn govern structural sizing.

6.2 Spanwise Lift Distribution

For analytical convenience and physical relevance, the lift distribution is assumed to follow an elliptical profile. The lift per unit span is expressed as:

$$w(y) = w_0 \sqrt{1 - \frac{4y^2}{b^2}}, \quad -\frac{b}{2} \leq y \leq \frac{b}{2} \tag{6.1}$$

The total lift over the wing is obtained by integrating the distribution:

$$L = \int_{-b/2}^{b/2} w(y)\, dy \tag{6.2}$$

Substituting the assumed form:

$$L = \int_{-b/2}^{b/2} w_0 \sqrt{1 - \frac{4y^2}{b^2}}\, dy = w_0 \frac{\pi b}{4} \tag{6.3}$$

Thus, the peak load intensity is:

$$w_0 = \frac{4L}{\pi b} \tag{6.4}$$

The final expression for the lift distribution becomes:

$$w(y) = \frac{4L}{\pi b} \sqrt{1 - \frac{4y^2}{b^2}} \tag{6.5}$$

For semi-span analysis ($0 \leq y \leq s$), the expression is equivalently written as:

$$w(y) = \frac{4L}{\pi b}\sqrt{1 - \frac{y^2}{s^2}} \tag{6.6}$$

6.3 Shear Force Distribution

The shear force at any spanwise location is obtained by integrating the distributed load from that location to the wing tip:

$$V(y) = \int_y^s w(\eta)\, d\eta \tag{6.7}$$

Substituting the lift distribution:

$$V(y) = \frac{4L}{\pi b}\int_y^s \sqrt{1 - \frac{\eta^2}{s^2}}\, d\eta \tag{6.8}$$

Using the substitution:

$$\eta = s\sin\theta$$

the integral simplifies to:

$$V(y) = \frac{L}{\pi}\left[\cos^{-1}\left(\frac{y}{s}\right) - \frac{y}{s}\sqrt{1 - \frac{y^2}{s^2}}\right] \tag{6.9}$$

Boundary Conditions:

- At root ($y = 0$): $V(0) = \frac{L}{2}$
- At tip ($y = s$): $V(s) = 0$

6.4 Bending Moment Distribution

The bending moment is obtained by integrating the shear force:

$$M(y) = \int_y^s V(\xi)\, d\xi \tag{6.10}$$

Alternatively, it can be expressed directly in terms of the load distribution:

$$M(y) = \int_y^s w(\xi)(\xi - y)\, d\xi \tag{6.11}$$

Evaluating the integral yields:

$$M(y) = \frac{bL}{3\pi}\left(1 - \frac{y^2}{s^2}\right)^{3/2} - yV(y) \tag{6.12}$$

At the root:

$$M(0) = \frac{bL}{3\pi} \tag{6.13}$$

This value represents the maximum bending moment and is critical for structural sizing.

6.5 Torsional Moment Distribution

The torsional moment arises due to the offset between the aerodynamic center and the shear center. Assuming a constant offset e, the torsional moment is:

$$T(y) = \int_y^s w(\xi)\, e\, d\xi \tag{6.14}$$

Since e is constant:

$$T(y) = e\, V(y) \tag{6.15}$$

At the root:

$$T(0) = e\frac{L}{2} \tag{6.16}$$

6.6 Summary of Load Distributions

The derived expressions provide complete spanwise distributions of:

- Lift per unit span $w(y)$
- Shear force $V(y)$
- Bending moment $M(y)$
- Torsional moment $T(y)$

These quantities form the input for structural analysis and sizing.

Chapter 7

Structural Analysis Methodology

7.1 Introduction

This chapter establishes the methodology used to convert aerodynamic loads into structural design parameters. The approach integrates beam theory and composite laminate mechanics to determine stiffness requirements and structural dimensions.

7.2 Bending Stiffness Requirement

A key design constraint is the maximum allowable deflection at the wing tip. Using beam theory, the tip deflection is expressed as:

$$\delta_{\text{tip}} = \frac{1}{EI} \int_0^s M(y)(s-y)\, dy \tag{7.1}$$

Rearranging, the required bending stiffness is:

$$EI = \frac{\int_0^s M(y)(s-y)\, dy}{\delta_{\text{tip}}} \tag{7.2}$$

This equation provides the target stiffness that the wing structure must achieve.

7.3 Composite Laminate Modeling

The wing structure is composed of composite laminates. Each lamina is characterized by an orthotropic stiffness matrix:

$$[Q] = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix} \tag{7.3}$$

For plies oriented at an angle, the stiffness matrix is transformed to obtain $[\bar{Q}]$.
The laminate stiffness matrices are obtained by integrating through the thickness:

$$A = \sum \bar{Q}_k (z_k - z_{k-1}) \tag{7.4}$$

$$B = \frac{1}{2} \sum \bar{Q}_k (z_k^2 - z_{k-1}^2) \tag{7.5}$$

$$D = \frac{1}{3} \sum \bar{Q}_k (z_k^3 - z_{k-1}^3) \tag{7.6}$$

These matrices collectively define the laminate response.

7.4 Effective Material Properties

For structural sizing, the laminate is approximated as an equivalent homogeneous material. The effective modulus is obtained from the compliance matrix:

$$S = A^{-1} \tag{7.7}$$

$$E_{\text{eff}} = \frac{1}{t \cdot S_{11}} \tag{7.8}$$

Separate values are obtained for spar caps and skins based on their laminate configurations.

7.5 Spar Cap Sizing Methodology

The spar caps are sized to satisfy the bending stiffness requirement. Using:

$$EI = E_{\text{cap}} \cdot I \tag{7.9}$$

the required second moment of area is:

$$I = \frac{EI}{E_{\text{cap}}} \tag{7.10}$$

This is translated into geometric dimensions based on the spacing between spar caps and assumed cross-sectional shape.

7.6 Torsional Stiffness Evaluation

The torsional rigidity of the wing box is evaluated using thin-walled closed section theory:

$$GJ = \frac{4A_m^2}{\sum \frac{s_i}{G_i t_i}} \tag{7.11}$$

where A_m is the enclosed area and the denominator accounts for shear deformation in each wall segment.

7.7 Rib Spacing Determination

The rib spacing is determined based on shear flow considerations. The shear flow is approximated as:

$$q = \frac{L}{2(h_f + h_r)} \tag{7.12}$$

The corresponding shear stress is:

$$\tau = \frac{q}{t_{\text{skin}}} \tag{7.13}$$

The rib spacing is then estimated as:

$$s_{\text{rib}} \propto t_{\text{skin}} \sqrt{\frac{E_{\text{skin}}}{\tau}} \tag{7.14}$$

This ensures that the skin panels remain stable under shear loading.

7.8 Summary

The methodology developed in this chapter provides a systematic framework for converting aerodynamic loads into structural design parameters. It establishes the link between load distributions, material properties, and geometric sizing of the wing-box structure.

Chapter 8

Results and Failure Modes

8.1 Overview

This chapter presents the structural results obtained from the wing-box sizing procedure and evaluates the behavior of the structure under bending loads. The analysis focuses on stiffness, load distribution, and failure characteristics of the composite laminate.

8.2 Wing-Box Structural Dimensions

The final dimensions obtained from the design are:

- Spar cap thickness: $t_{\text{sparcap}} = 0.004\,\text{m}$
- Spar web thickness: $t_{\text{sparweb}} = 0.004\,\text{m}$
- Skin thickness: $t_{\text{skin}} = 0.002\,\text{m}$
- Front spar cap width: $0.2747\,\text{m}$
- Rear spar cap width: $0.1831\,\text{m}$
- Rib spacing: $0.1080\,\text{m}$

Compared to preliminary estimates, the spar caps are significantly wider, indicating that bending stiffness is the dominant design requirement.

8.3 Global Structural Properties

The computed structural properties of the wing-box are:

- Bending stiffness: $EI = 1.739 \times 10^6\,\text{N} \cdot \text{m}^2$
- Torsional rigidity: $GJ = 5.897 \times 10^5\,\text{N} \cdot \text{m}^2$

- Spar cap modulus: $E_{\text{sparcap}} = 1.12 \times 10^{11}\,\text{Pa}$
- Skin modulus: $E_{\text{skin}} = 1.69 \times 10^{10}\,\text{Pa}$

The increased bending stiffness reflects the stricter tip deflection constraint used in this design iteration.

8.4 Load Distribution Plots

The spanwise distributions of lift, shear force, and bending moment are shown below.

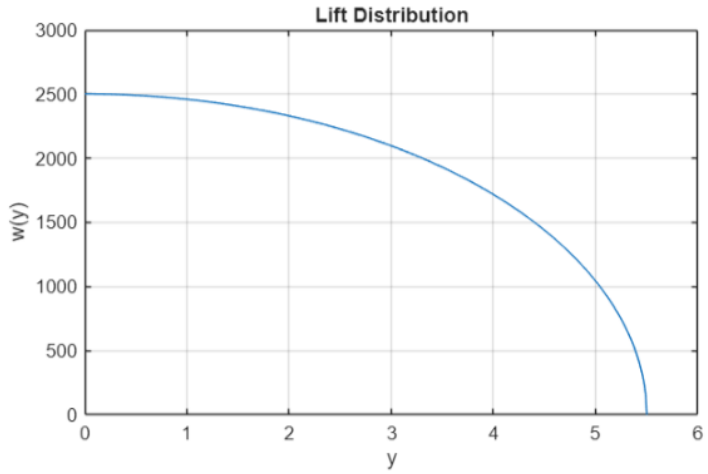


Figure 8.1: Spanwise lift distribution

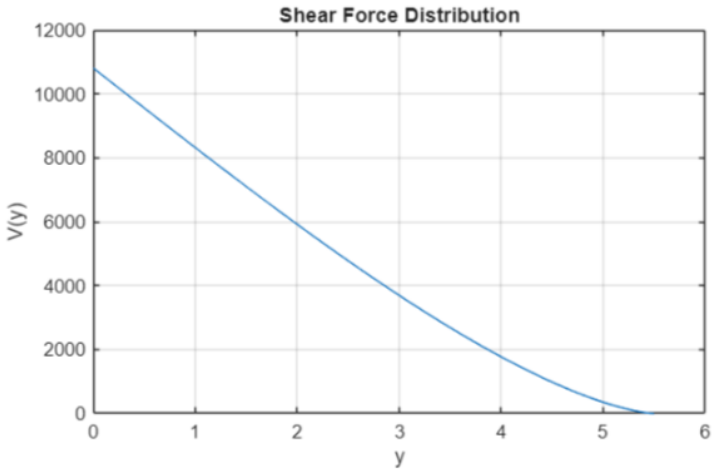


Figure 8.2: Shear force distribution along the span

As observed:

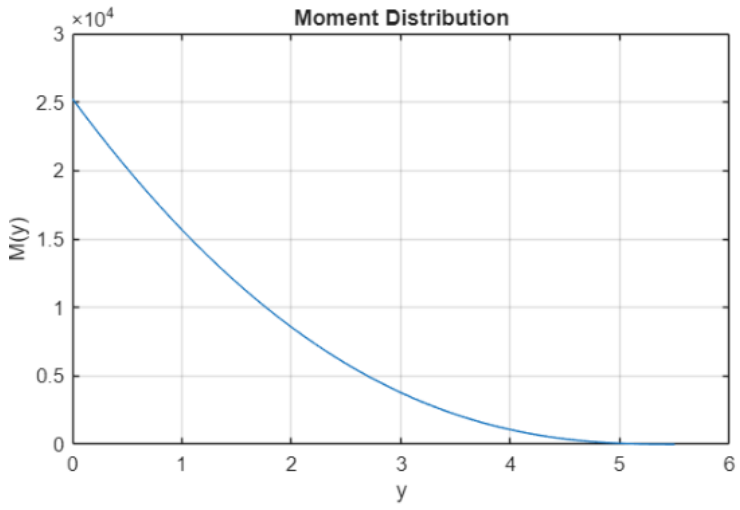


Figure 8.3: Bending moment distribution along the span

- Lift follows an elliptical distribution with maximum at the root.
- Shear force decreases monotonically from root to tip.
- Bending moment is maximum at the root and reduces to zero at the tip.

8.5 Critical Load Conditions

At the wing root, the governing loads are:

$$V_{\text{root}} = 1.081 \times 10^4 \,\text{N}$$
$$M_{\text{root}} = 2.523 \times 10^4 \,\text{N} \cdot \text{m}$$

These values confirm that the structure is primarily governed by bending loads.

8.6 Failure Modes under Bending

The wing-box is subjected predominantly to bending, leading to the following failure mechanisms:

8.6.1 Fiber-Dominated Failure

The primary failure mode is axial failure in the spar caps due to bending-induced stresses. The upper spar cap experiences compression, while the lower spar cap experiences tension.

8.6.2 Matrix Failure

Matrix-dominated failure may occur in off-axis plies due to transverse stresses. However, its contribution is secondary compared to fiber failure.

8.6.3 Shear Failure

Shear stresses are mainly carried by the spar webs and skins. Due to sufficient thickness and material properties, shear failure is not expected to govern the design.

8.7 Discussion

The results indicate that:

- The structure is bending-dominated, with maximum stresses at the root.
- Increasing spar cap width significantly enhances bending stiffness.
- The torsional rigidity is sufficiently high due to the closed-section wing-box.
- Reduced rib spacing improves shear stability of the skin panels.

Overall, the design satisfies stiffness and strength requirements under limit loading conditions.

Chapter 9

Conclusions

9.1 Summary of Work

This study developed a preliminary structural design of a composite wing-box using analytical methods. The approach combined aerodynamic load modeling, beam theory, and Classical Laminate Theory to determine structural dimensions and performance.

9.2 Key Findings

- The wing structure is governed primarily by bending loads, with maximum stresses occurring at the root.
- The bending stiffness requirement drives the sizing of spar caps, resulting in relatively large spar cap widths.
- The torsional rigidity of the wing-box is sufficiently high due to the closed-section configuration formed by skins and spars.
- The use of composite laminates enables tailoring of material properties to achieve the required stiffness.
- The chosen rib spacing ensures adequate resistance to shear-induced instability.

9.3 Failure Mode Insights

The dominant failure mechanism is fiber-direction failure in the spar caps due to bending-induced axial stresses. Matrix and shear failures are secondary and do not govern the design.

This confirms that the structural design is controlled by bending stiffness and strength requirements.

9.4 Effect of Design Choices

Compared to earlier iterations, the reduced allowable tip deflection resulted in:

- Increased bending stiffness requirement
- Larger spar cap dimensions
- Higher overall structural rigidity

This highlights the sensitivity of structural design to deflection constraints.

9.5 Limitations

The analysis is based on simplified assumptions:

- Elliptical lift distribution
- Linear elastic material behavior
- Neglect of local buckling and stress concentrations
- No aeroelastic coupling effects

These assumptions are appropriate for preliminary design but require refinement in detailed analysis.

9.6 Future Work

Future improvements may include:

- Buckling analysis of skin panels and spar webs
- Advanced composite failure criteria (Tsai-Wu, Hashin)
- Finite element modeling for validation
- Aeroelastic analysis for coupled behavior
- Optimization of laminate stacking sequence

9.7 Final Remarks

The present work demonstrates a consistent analytical framework for preliminary composite wing design. The results provide a strong foundation for further refinement and detailed structural analysis.

ABBD Matrix Calculation Code

Datasheet Syntax

Row 1: Thickness <value>

Row 2 : numLayers <value>

Row 3: Temperature (T) <value>

Row 4: Moisture (M) <value>

Row 5 : --blank--

Row 6 : (Headers) Thickness Angle E1 E2 nu12 nu21 G12 alpha_xx alpha_xy alpha_yy Tref b_xx b_xy b_yy Mref

Row 7+: Lamina parameter values as per row 4

Q Matrix Calculator (Stiffness Matrix of a lamina)

Input: E1 E2 nu12 nu21 G12

Output: Q (or C)

```
function [Q] = Q_matrix_calculator(E1, E2, nu12, nu21, G12)
    Q = zeros(3, 3);
    Q(1, 1) = (-E1)/(nu12*nu21 - 1);
    Q(1, 2) = (-nu12*E2)/(nu12*nu21 - 1);
    Q(1, 3) = 0;
    Q(2, 1) = (-nu21*E1)/(nu12*nu21 - 1);
    Q(2, 2) = (-E2)/(nu12*nu21 - 1);
    Q(2, 3) = 0;
    Q(3, 1) = 0;
    Q(3, 2) = 0;
    Q(3, 3) = G12;
end
```

Qbar Matrix Calculator (Transformed Stiffness Matrix of a lamina)

Input: Q theta

Output: Qbar

```
function [Qbar] = Qbar_matrix_calculator(Q, theta)
    Qbar = zeros(3, 3);
    m = cos(deg2rad(theta));
    n = sin(deg2rad(theta));

    Qbar(1, 1) = Q(1, 1)*(m^4) + 2*(Q(1, 2) + 2*Q(3, 3))*(n^2)*(m^2) + Q(2,
2)*(n^4);
    Qbar(1, 2) = (Q(1, 1) + Q(2, 2) - 4*Q(3, 3))*(n^2)*(m^2) + Q(1, 2)*(n^4 + m^4);
```

```
    Qbar(1, 3) = (Q(1, 1) - Q(1, 2) - 2*Q(3, 3))*n*(m^3) + (Q(1, 2) - Q(2, 2) +
2*Q(3, 3))*(n^3)*m;
    Qbar(2, 1) = Qbar(1, 2);
    Qbar(2, 2) = Q(1, 1)*(n^4) + 2*(Q(1, 2) + 2*Q(3, 3))*(n^2)*(m^2) + Q(2,
2)*(m^4);
    Qbar(2, 3) = (Q(1, 1) - Q(1, 2) - 2*Q(3, 3))*(n^3)*m + (Q(1, 2) - Q(2, 2) +
2*Q(3, 3))*n*(m^3);
    Qbar(3, 1) = Qbar(1, 3);
    Qbar(3, 2) = Qbar(2, 3);
    Qbar(3, 3) = (Q(1, 1) + Q(2, 2) - 2*Q(1, 2) - 2*Q(3, 3))*(n^2)*(m^2) + Q(3,
3)*(n^4 + m^4);
end
```

A, B, D Matrices Calculator Code

Input: Location of the csv file of composite parameters (as defined above)

Output: 6x6 Composite Stiffness Matrix [A, B; B, D]

```
function [composite_stiffness] = ABBD_matrix_calculator (path)
    % Extracting data from path
    data = readcell(path);

    % Initialisation of A, B, D and final composite matrices
    A = zeros(3, 3);
    B = zeros(3, 3);
    D = zeros(3, 3);
    composite_stiffness = zeros(6, 6);

    %Extracting composite parameters
    thickness = data{1, 2};
    numlayers = data{2, 2};

    z_tillnow = -thickness/2;

    for k = 1:numlayers
        % Extracting layer information
        layer_thickness = data{6 + k, 1};
        theta = data{6 + k, 2};
        E1 = data{6 + k, 3};
        E2 = data{6 + k, 4};
        nu12 = data{6 + k, 5};
        nu21 = data{6 + k, 6};
        G12 = data{6 + k, 7};

        Q = Q_matrix_calculator(E1, E2, nu12, nu21, G12);
        Qbar = Qbar_matrix_calculator(Q, theta);

        % Adding layer contribution to A, B, D matrices
        for i = 1:3
            for j = 1:3
```

```
                A(i, j) = A(i, j) + Qbar(i, j)*(layer_thickness);
                B(i, j) = B(i, j) + (0.5) * Qbar(i, j)*((z_tillnow +
layer_thickness)^2 - z_tillnow^2);
                D(i, j) = D(i, j) + (1/3) * Qbar(i, j)*((z_tillnow +
layer_thickness)^3 - z_tillnow^3);
            end
        end

        z_tillnow = z_tillnow + layer_thickness;
    end

    % Defining Composite stiffness matrix
        composite_stiffness(1:3, 1:3) = A;
        composite_stiffness(1:3, 4:6) = B;
        composite_stiffness(4:6, 1:3) = B;
        composite_stiffness(4:6, 4:6) = D;
end
```

Wing Box Analysis

```
% Given Parameters

MTOW = 580; % Maximum TakeOff Weight (kg)
g = 9.81; % gravitational acceleration (m/s^2)
W = MTOW * g; % Weight of aircraft (N)
n_limit = 3.8; % Limit Load Factor
n_ultimate = 1.5 * n_limit; % Ultimate Load Factor
delta_tip = 0.1; % Maximum allowed deflection at wing tip at limit load (m)

b = 11; % Wing span (m)
s = b/2; % Semi wing span (m)
c = 1.23; % Chord length (m)
NACA_number = 2412; % NACA Airfoil number

x_frontSpar = 0.25*c;
x_rearSpar = 0.65*c;



% ABBD Matrices from Composites

M_sparcap =
ABBD_matrix_calculator("C:\Users\anutt\OneDrive\Documents\Courses\AE673\Project\data
sheet_sparcap.xlsx");
M_skin =
ABBD_matrix_calculator("C:\Users\anutt\OneDrive\Documents\Courses\AE673\Project\data
sheet_skin.xlsx");
M_sparweb =
ABBD_matrix_calculator("C:\Users\anutt\OneDrive\Documents\Courses\AE673\Project\data
sheet_sparweb.xlsx");
```

```
data_sparcap =
readcell("C:\Users\anutt\OneDrive\Documents\Courses\AE673\Project\datasheet_sparcap.
xlsx");
data_skin =
readcell("C:\Users\anutt\OneDrive\Documents\Courses\AE673\Project\datasheet_skin.xls
x");
data_sparweb =
readcell("C:\Users\anutt\OneDrive\Documents\Courses\AE673\Project\datasheet_sparweb.
xlsx");



% Helper Functions

function [wf, wr] = SparwebWidth_Calculator (NACA_number, c, xf, xr)
    t = (mod(NACA_number, 100) / 100) * c;
    y_t = @(x) 5*t*(0.2969*sqrt(x) - 0.126*x - 0.3516*x^2 + 0.2843*x^3 -
0.1015*x^4);
    wf = 2 * y_t(xf/c);
    wr = 2 * y_t(xr/c);
end

function [EI] = EI_calculator (M, delta_tip, s)
    integrand = @(y) M(y) .* (s - y);
    K = integral(integrand, 0, s);
    EI = K / delta_tip;
end

function [E_sparcap] = E_sparcap_Calculator (M_sparcap, t_sparcap)
    A_sparcap = M_sparcap(1:3, 1:3);
    S_sparcap = inv(A_sparcap);
    E_sparcap = 1 / (t_sparcap * S_sparcap(1, 1));
end

function [wf, wr] = SparcapWidth_Calculator (EI, E_sparcap, width_frontSparweb,
width_rearSparweb, t_sparcap)
    I_spar = EI/E_sparcap;
    total_width_sparcap = I_spar / ((0.3 * width_frontSparweb^2 + 0.2 *
width_rearSparweb^2) * t_sparcap);
    wf = 0.6 * total_width_sparcap;
    wr = 0.4 * total_width_sparcap;
end

function [GJ] = GJ_Calculator (hf, hr, rib_length, A66_skin, A66_sparweb)
    Am = 0.5 * (hf + hr) * rib_length;
    val = (2 * rib_length) / A66_skin + (hf + hr) / A66_sparweb;
    GJ = (4*Am^2)/val;
end
```

```
function [E_skin] = E_skin_Calculator (M_skin, t_skin)
    A_skin = M_skin(1:3,1:3);   % N/m
    S_skin = inv(A_skin);       % compliance matrix
    E_skin = 1 / (t_skin * S_skin(1,1));   % Pa
end

function [ribs_spacing] = RibsSpacing_Calculator (L, hf, hr, t_skin, E_skin)
    q = (0.5*L)/(hf + hr);
    tau = q/t_skin;
    ribs_spacing = 5 * t_skin * sqrt(E_skin / tau);
end



% Derived Quantities

t_sparcap = data_sparcap{1,2};
t_skin = data_skin{1,2};
t_sparweb = data_sparweb{1,2};

[width_frontSparweb, width_rearSparweb] = SparwebWidth_Calculator(NACA_number, c,
x_frontSpar, x_rearSpar);
rib_length = x_rearSpar - x_frontSpar;

L_limit = n_limit * W; % Lift at limit load (N)
L_ultimate = n_ultimate * W; % Lift at ultimate load (N)

V = @(y) (L_limit/pi) * (acos(y/s) - (y/s).*sqrt(1 - (y/s).^2));
M = @(y) ((b*L_limit)/(3*pi)) * (1 - (y/s).^2).^1.5 - y.*V(y);

EI = EI_calculator(M, delta_tip, s); % Bending Stiffness (N/m^2)

E_sparcap = E_sparcap_Calculator(M_sparcap, t_sparcap);

[width_frontSparcap, width_rearSparcap] = SparcapWidth_Calculator (EI, E_sparcap,
width_frontSparweb, width_rearSparweb, t_sparcap);

GJ = GJ_Calculator(width_frontSparcap, width_rearSparcap, rib_length, M_skin(3, 3),
M_sparweb(3, 3));

E_skin = E_skin_Calculator(M_skin, t_skin);

ribs_spacing = RibsSpacing_Calculator(L_limit, width_frontSparcap,
width_rearSparcap, t_skin, E_skin);


% Printing Results

disp("Data regarding wingbox dimension: --");
```

```
Data regarding wingbox dimension: --
```

```
t_sparcap
```

```
t_sparcap =
0.0040
```

```
t_sparweb
```

```
t_sparweb =
0.0040
```

```
t_skin
```

```
t_skin =
0.0020
```

```
width_frontSparcap
```

```
width_frontSparcap =
0.2747
```

```
width_rearSparcap
```

```
width_rearSparcap =
0.1831
```

```
rib_length
```

```
rib_length =
0.4920
```

```
rib_spacing
```

```
rib_spacing =
0.1080
```

```
disp("------------------------------")
```

```
------------------------------
```

```
disp("Property Data: --");
```

```
Property Data: --
```

```
L_limit
```

```
L_limit =
2.1621e+04
```

```
EI
```

```
EI =
1.7394e+06
```

```
GJ
```

```
GJ =
```

```
5.8972e+05
```

```
E_sparcap
```

```
E_sparcap =
1.1207e+11
```

```
E_skin
```

```
E_skin =
1.6911e+10
```

```
disp("------------------------------")
```

```
------------------------------
```

```
disp("At root values: --");
```

```
At root values: --
```

```
V_root = V(0)
```

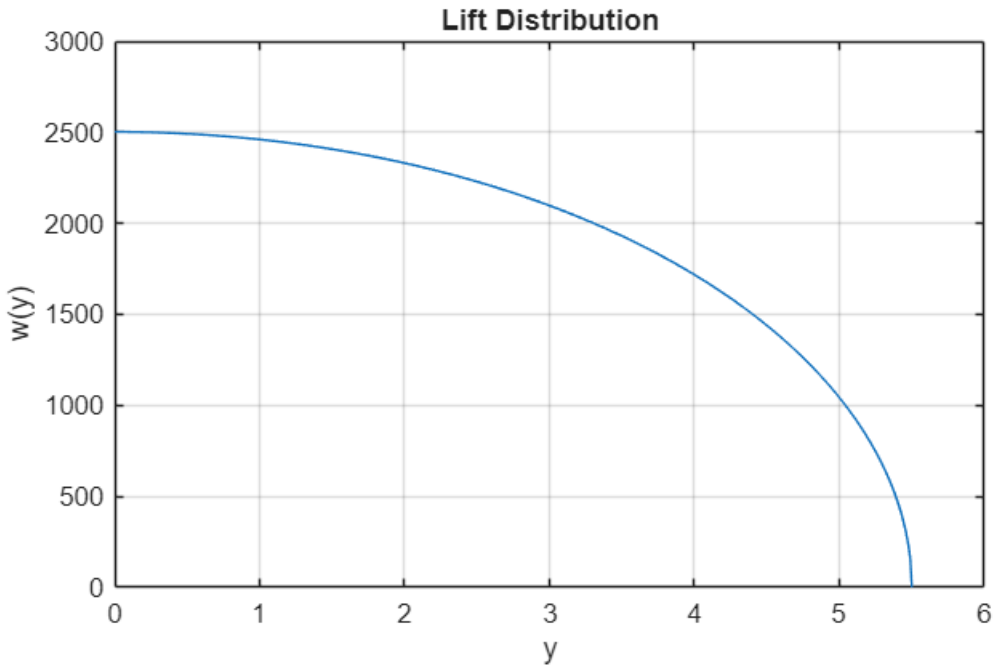
```
V_root =
1.0811e+04
```

```
M_root = M(0)
```

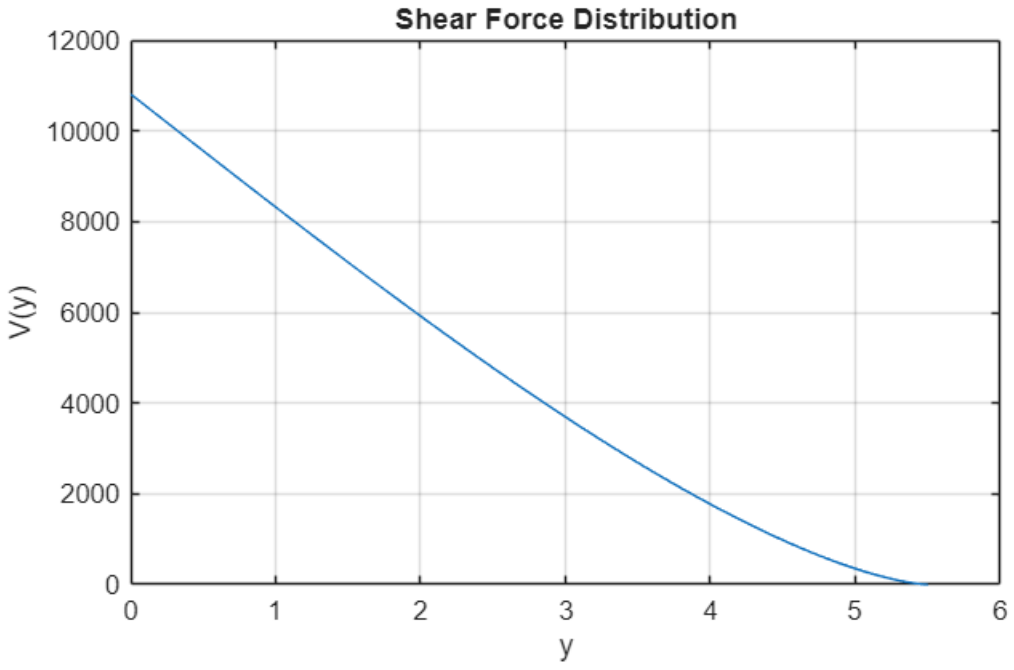
```
M_root =
2.5235e+04
```

```
% Distribution Plots
y = 0:1e-2:s;

figure;
plot(y, ((4*L_limit)/(pi*b)) * sqrt(1 - y.^2/s^2));
xlabel("y");
ylabel("w(y)");
title("Lift Distribution");
grid on;
```




```
figure;
plot(y, V(y));
xlabel("y");
ylabel("V(y)");
title("Shear Force Distribution");
grid on;
```




```
figure;
plot(y, M(y));
```

```
xlabel("y");
ylabel("M(y)");
title("Moment Distribution");
grid on;
```

